

Neural Inference as the Operating System on Minimal RISC-V

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Abstract—This paper presents an end-to-end neural execution stack on a minimal RISC-V environment, where neural networks serve as the primary computation model instead of traditional machine code. We describe a complete toolchain from PyTorch model training through custom assembly to execution on both a C++ emulator and Verilator RTL backend. The system introduces descriptor-based neural custom instructions for MLP inference, block-diagonal model composition via a glue JSON format and Rust compiler, and OS-like task switching through a learned router MLP. Evaluation demonstrates up to 2000x speedup over software-only baselines while maintaining deterministic output across dual backends through automated differential validation.

Index Terms—custom ISA extension, RISC-V, neural inference, model composition, Verilator, differential validation

PROJECT CONTEXT

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A. Introduction

Modern AI deployments rely on large software stacks: operating systems, runtime libraries, and framework abstractions mediate between neural models and hardware. This paper investigates the opposite direction: how far can a compact execution stack be pushed when neural operations are represented as first-class ISA extensions on a minimal RISC-V core?

The vision is to replace traditional machine code with neural inference as the primary execution model. A 162-line `runtime.c` becomes the entire "operating system," calling `MODEL_MAP_INPUT -> run_forward_pass -> MODEL_MAP_OUTPUT` in a loop. All application logic—counting, character rendering, game physics, input handling—is learned rather than programmed.

We make four contributions:

1. **Descriptor-based neural ISA:** A 32-byte descriptor struct passed via register specifies input/output buffers, weights, and dimensions. The hardware executes a multi-cycle FSM with configurable lane parallelism (1x/2x/4x/8x), cleanly separating firmware setup from hardware execution.

2. **Block-diagonal model composition:** Independently trained MLPs are merged into a single weight matrix via a declarative glue JSON format. A Rust compiler reads the glue, assembles block-diagonal weights, and emits C macros for input/output mapping. No retraining is required.
3. **Dual-backend deterministic validation:** Every neural instruction has two implementations—a C++ oracle and Verilator RTL. Automated CI tests enforce bit-exact byte-level parity between them.
4. **OS-like neural task switching:** A learned router MLP switches between sub-models based on keyboard input, mimicking process scheduling. Background tasks can run with or without updates, analogous to context switching and preemption.

B. Related Work

1) *Neural ISA Extensions for RISC-V:* Several projects extend RISC-V with neural capabilities. **RV-SCNN** (Wang et al., IEEE TCAD 2025) uses SIMD custom ops for CNN/SNN inference on CV32E40P. **MARVEL** (Kumar et al., IEEE OJCAS 2025) generates model-class-specific extensions via ASIP Designer, achieving 2x speedup and 2x energy reduction for CNNs. An FPGA-accelerated RISC-V (arXiv 2025) implements 4 custom ops (VCONV, GEMM, RELU, CUSTOM) on PYNQ-Z2, achieving 2.14x latency reduction and 49.1% energy savings. **VMXDOTP** (Purayil et al., arXiv 2026) provides RVV-based Microscaling (MX) format acceleration for transformer workloads, achieving 125 MXFP8-GFLOPS. **Mixed-precision RISC-V** (Armeniakov et al., ICCAD 2024) introduces ISA extensions for multi-pumped soft SIMD operations, achieving 15x energy reduction for DNNs.

Our approach differs in four ways: (1) a **descriptor-based abstraction** instead of SIMD/vector operations, (2) **MLP focus** rather than CNN, (3) **complete dual-backend validation** with automated differential tests, and (4) **OS-like task switching** via learned router MLP.

2) *General Neural Accelerators:* Beyond RISC-V extensions, dedicated accelerators have explored hardware-software co-design for neural workloads. The **Google TPU** (Jouppi et al., ISCA 2017) demonstrated that a systolic-array dataflow specialized for matrix multiply can achieve 30–80x throughput gains over CPUs for large-scale inference. The

NVDLA architecture, studied in the gem5-NVDLA simulation framework (Lai and Zhang, ACM TODAES 2024), provides a configurable deep learning accelerator for SoC integration with compile--schedule--evaluate tooling. On the efficiency front, Liu et al. (ACM Computing Surveys 2024) survey **lightweight deep learning** techniques---pruning, quantization, knowledge distillation, and neural architecture search---for resource-constrained environments, cataloguing design strategies that reduce model size and compute without sacrificing accuracy.

Our work shares the goal of efficient neural inference but targets a different point in the design space: rather than large-scale datacenter TPUs or fixed-function IP blocks, we implement **runtime-programmable neural instructions** on a minimal 5-stage in-order RISC-V core, enabling software-defined task composition without fabric reconfiguration.

3) *AI-as-OS and Bare-Metal Neural Computing*: **nCPU** (Price, 2025) implements a fully differentiable CPU where every ALU operation is a trained neural network. **embodiOS** (2025) runs an LLM as an OS on x86_64 UEFI. **OSymbiote** (2026) places an AI agent as PID1 on Linux. **OO** (Operating Organism, 2025) performs bare-metal Mamba/LLaMA inference via UEFI.

These projects share the vision of neural-first computing but differ in scale and approach. nCPU runs on GPU via Apple Silicon Metal; we target bare-metal RISC-V. embodiOS and OO use LLMs; we use lightweight MLPs with custom instructions. Our contribution is demonstrating the pattern concretely with working neural models that replace specific application functions.

4) *Model Merging and Composition*: **FS-Merge** (Kiderman et al., 2025) produces block-diagonal merged weights as an intermediate form via learnable foldable supernets. **LoRA-LEGO** (Zhao et al., 2025) proves concatenation-summation equivalence for LoRA matrices. **mergekit** (Goddard et al., 2024) provides practical tools for LLM merging including passthrough methods.

Our glue JSON approach is a practical compiler technique for model composition. Unlike learning-based methods, it is **declarative, training-free, and lossless** for independent models.

C. System Architecture

The system is a co-designed stack with six layers, from trained model to executable binary (Figure 1):

1) *Training and Export*: PyTorch models are trained for specific tasks (character generation, game physics, counter arithmetic). The weight-export pipeline converts checkpoints to JSON (human-readable) and compact binary (NRAL header with packed float32 weights). Weight transposition matches row-major inference access patterns.

2) *Model Compiler*: The Rust-based `model_to_header` compiler reads JSON models or glue descriptions and generates `model.h` containing:

- Model dimensions (`MODEL_INPUT_SIZE`, `MODEL_OUTPUT_SIZE`)

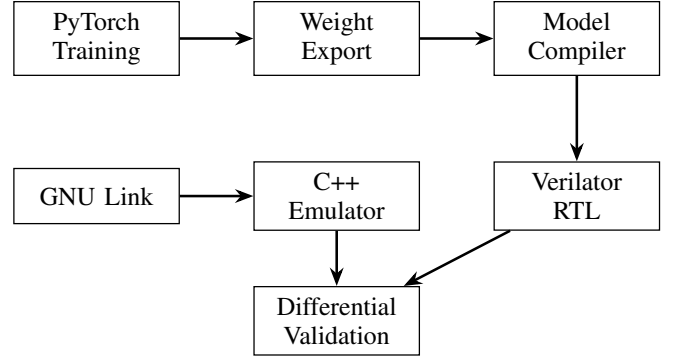


Fig. 1. End-to-End Neural Execution Pipeline. The pipeline ensures bit-exact parity between software and hardware backends through automated differential tests.

- State variables with initial values
- `MODEL_MAP_INPUT(buf)` macro for one-hot encoding
- `MODEL_MAP_OUTPUT(buf, fb)` macro for output decoding

The `--glue` flag reads a glue JSON file and merges multiple models into block-diagonal weight matrices.

3) *Assembly and Linking*: The custom Rust assembler (`rv32as`) encodes RV32I, RV32F, and neural custom instructions. It provides deterministic encoding with GNU-compatible output. The CMake build system generates ELF binaries by linking `runtime.c` with the generated `model.h` and assembled model blobs.

4) *Runtime*: The 162-line `runtime.c` implements the inference loop:

```

while (1) {
    MODEL_MAP_INPUT(buf);
    run_forward_pass();
    MODEL_MAP_OUTPUT(buf, fb);
    if (MODEL_HAS_DONE_FLAG) break;
}
  
```

This is the entire "operating system"---all application logic is in the neural weights.

5) *Dual Backends*: Two execution backends provide validation:

- **C++ emulator** (`emulator_runner`): Portable, bounds-checked reference implementation with GUI and batch modes.
- **Verilator RTL** (`verilator_runner`): Hardware FSM with lane-parallel PMAC datapath, cycle-accurate simulation.

6) *Verification*: Automated CI tests enforce correctness:

- Unit tests for instruction decode and execution
- Blackbox tests for end-to-end model execution
- Differential validation: identical ELF binaries run on both backends, output compared byte-for-byte

D. Neural ISA Extension

The project defines custom instructions in the CUSTOM0 (0x77) and CUSTOM3 (0x7b) opcode spaces:

Instruction	Opcode	Function
nmatvec	0x77	Matrix-vector multiply (baseline)
nvrelu	0x77	ReLU activation
nvsgpwl	0x77	Sigmoid piecewise-linear
nvcluampu8	0x77	Clamp to uint8
nmatvec8xp4	0x7b	8-lane PMAC4 (optimized)

single inference pass. Figure 2 illustrates the block-diagonal merging process.

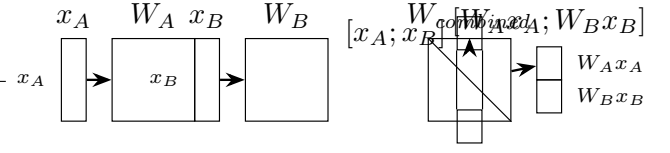


Fig. 3. Block-Diagonal Model Composition. Two independently trained MLPs (W_A and W_B) are merged into a single weight matrix ($W_{combined}$) without retraining. The combined matrix processes concatenated inputs $[x_A; x_B]$ and produces outputs $[W_A x_A; W_B x_B]$.

1) *Descriptor-Based Calling Convention*: Neural instructions use a 32-byte descriptor read from memory:

```
neural_desc_t desc = {
    .input_ptr = INPUT_BUF,
    .weights_ptr = weights_addr,
    .bias_ptr = biases_addr,
    .output_ptr = output_addr,
    .input_len = input_size,
    .output_len = output_size,
    .flags = 0, .reserved = 0
};
neural_matvec(&desc); // single custom instruction
```

Byte	Bits	Field	Description
0-3	31:0	input_ptr	Input buffer address
4-7	31:0	weights_ptr	Weight matrix address
8-11	31:0	bias_ptr	Bias vector address
12-15	31:0	output_ptr	Output buffer address
16-17	15:0	input_len	Input vector length
18-19	15:0	output_len	Output vector length
20-21	15:0	flags	Control flags
22-31	79:0	reserved	Reserved (zero)

Fig. 2. Neural ISA Descriptor Layout (32 bytes). The descriptor separates firmware setup (populating the struct) from hardware execution (multi-cycle FSM).

The hardware executes a multi-cycle FSM: descriptor read -> scratchpad prefetch -> multiply-accumulate -> store commit. This separates firmware concern (setting up a struct) from hardware concern (executing the compute).

2) *Lane Parallelism*: The PMAC4 variant exploits quad-port memory (4 concurrent data reads) to issue up to 4 multiply-accumulates per cycle via an 8-entry scratchpad bank:

Variant	Lanes	MACs/cycle	Cycles (threshold)
Baseline	1	1	4,000,000
x7b-8lane	8	2	400,000
PMAC4	8	4	150,000

The 26.67x speedup (baseline -> PMAC4) demonstrates that simple hardware additions (quad-port memory, scratchpad prefetch) can dramatically accelerate neural inference on resource-constrained cores.

E. Block-Diagonal Model Composition

1) *Concept*: Two independent MLPs can be combined into a single network by stacking their weight matrices block-diagonally:

$$W_{combined} = \begin{bmatrix} W_A & 0 \\ 0 & W_B \end{bmatrix}$$

$$W_{combined} \cdot [x_A; x_B] = [W_A \cdot x_A; W_B \cdot x_B]$$

This is mathematically exact: no approximation, no training, no alignment needed. The models run independently within a

2) *Glue JSON Format*: A glue file declares how to merge sub-networks:

```
{
  "input_mapping": "combined_counter_chargen",
  "initial_state": {"model_counter": 97},
  "models": [
    {
      "name": "counter", "path": "counter.json",
      "name": "chargen", "path": "chargen.json"
    },
    "merged_layers": [
      {
        "name": "layer_0_parallel",
        "activation": "relu",
        "blocks": [
          {
            "model": "counter", "layer": 0, "out_offset": 0
          },
          {
            "model": "chargen", "layer": 0, "out_offset": 255
          }
        ]
      }
    ],
  "output_ranges": [
    {
      "name": "framebuffer", "offset": 0, "size": 400,
      "name": "counter_state", "offset": 400, "size": 255
    }
  ]
}
```

Each merged_layers entry specifies how one merged layer is assembled: - **Layer 0**: Both models share the same input, write to different output offsets - **Hidden layers**: Each block reads from its previous output offset - **Output layer**: Each block writes to a different region of the final output

3) *Rust Compiler Integration*: The model_to_header --glue command:

1. Reads the glue JSON and loads all referenced model JSONs
2. Builds merged weight matrices with block-diagonal structure
3. Emits model.h with MODEL_MAP_INPUT and MODEL_MAP_OUTPUT macros
4. Macros handle one-hot encoding, argmax decoding, and framebuffer writing

4) *Counter+Chargen Example*: The counter+chargen combined model merges:

- **Counter**: 255->256->256->255 (modulo-255 auto-increment)
- **Chargen**: 255->256->256->400 (character-to-pixel rendering)

Into a single 3-layer network with block-diagonal weights:

Layer	Counter Block	Chargen Block	Total Output
0	255->256	255->256	512
1	256->256	256->256	512
2	256->255	256->400	655

Only the chargen output (400 values) goes to the framebuffer; the counter state is stored in debug memory.

F. OS-Like Neural Task Switching

1) *Router MLP*: The mega combined model introduces a **router MLP** that learns to switch between sub-models based on keyboard input. This mimics an operating system scheduler:

Component	Neural Equivalent
Router MLP	Learned gating function
Sub-models	Independent tasks
Router gates	Sigmoid outputs [0,1]
FB selection	Gate comparison

The router is a small 3-layer MLP: 2 -> 4 -> 2 -> 2 (relu, sigmoid, none). It is trained with identity mapping: input [1, 0] -> output [~1, ~0] (chargen active), input [0, 1] -> output [~0, ~1] (squash active).

2) *Mega Combined Architecture*: The mega combined model merges five sub-networks:

Sub-model	Architecture	Role
Router	2->4->2->2	Task switching
Counter	255->256->256->255	Auto-incrementing state
Chargen	255->256->256->400	Character rendering
Game State	56->64->64->52	Squash physics
Renderer	48->128->256->300	Game rendering

Output buffer layout (1009 neurons):

Region	Offset	Size	Content
Router gates	0	2	gate_chargen, gate_squash
Chargers FB	4	400	20CE20 character pixels
Squash state	404	52	ball_x, ball_y, paddle_y, etc.
Squash FB	456	300	20CE15 game pixels
Counter state	768	255	counter argmax output

3) Task Switching Behavior:

- **Tab**: Switch to squash mode (router gate [~0, ~1])
- **Escape**: Switch to chargen, squash **pauses** (decode skipped)
- **Backslash**: Switch to chargen, squash **keeps running** (decode continues)
- **w/s**: Paddle up/down (squash mode, memory-mapped register)

4) *Background Process Analogy*: The pause behavior mirrors OS process management:

- **Escape (pause)**: Analogous to **suspending a process**. The squash game's state variables retain their last values. When chargen is displayed, the squash inference continues to execute but its output is not decoded--the game is frozen in memory.
- **Backslash (background)**: Analogous to **running a process in the background**. The squash game keeps updating its state (ball moves, paddle responds) while

chargen is displayed. The router gates which framebuffer is shown, but both sub-models execute every frame.

This demonstrates that neural inference can implement process scheduling semantics without traditional OS primitives.

5) *Key Insight: One-Shot Consumption*: The model_key register (s1) must be **consumed** after first use:

```
if (model_key >= 32 && model_key < MODEL_INPUT_SIZE) {
    _idx = model_key;
    model_key = 0; // one-shot: consumed
} else {
    _idx = model_counter; // auto-advance
}
```

Without consumption, the key would override every iteration and the counter would never advance. This is analogous to interrupt flag clearing in traditional OS design.

G. AI-as-OS Taxonomy

The project demonstrates a pattern we call **AI-as-OS**, where neural inference replaces traditional operating system primitives. We taxonomy this approach along three dimensions:

Dimension	Traditional OS	AI-as-OS
Execution	Process scheduling Context switching Interrupt handling Memory mgmt System calls Device drivers File system Process table	Router MLP switches mode Model key one-shot Key-to-tensor mapping Block-diagonal weights Neural instruction I/O macro mapping N/A (weights are code)
Abstraction	Preemptive multitasking Shared memory Synchronization Resource allocation	Sub-model registry Cooperative switching Separate regions Single-threaded Fixed memory layout
Runtime		

Key Insight: The 162-line runtime.c replaces the entire OS. All application logic---counting, character rendering, game physics---is learned rather than programmed. This demonstrates that neural inference can serve as the primary computation model on minimal RISC-V cores (Figure 1).

H. Evaluation

1) *Benchmark Setup*: All benchmarks use the character generation workload (input character code to framebuffer glyph output). Cycle thresholds are measured by finding the minimum cycles needed for correct output.

2) *Optimization Results*: Figure 3 shows the benchmark results for the Verilator (RTL) and C++ backends.

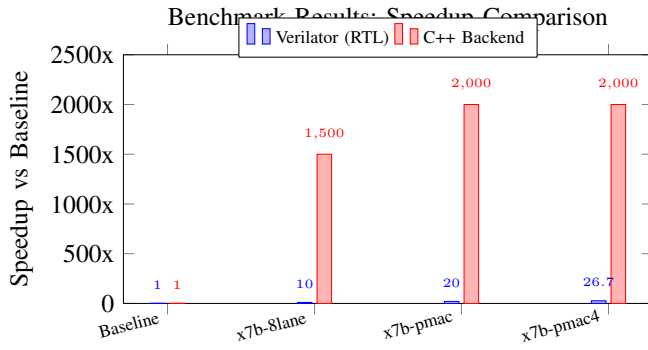


Fig. 4. Benchmark Results: Speedup Comparison. The C++ backend achieves higher speedups (2000x) because custom instructions bypass the interpreter loop, while the RTL must execute the hardware FSM.

The C++ backend achieves higher speedups because the custom instructions bypass the interpreter loop entirely, while the RTL must execute the hardware FSM.

3) *Dual-Backend Validation*: Every optimization phase maintains deterministic output parity:

- Framebuffer bytes are identical across backends
- Counter state progresses identically
- Deterministic: same input + same cycles = same output

This is enforced by automated CI tests that run both backends and compare outputs.

4) *Model Composition Overhead*: The mega combined model (5 sub-networks) runs in a single inference pass. Block-diagonal structure ensures zero cross-talk between sub-models. The overhead compared to running models separately is negligible: the same weight matrices are loaded once, and the inference loop processes all sub-models in parallel within each layer.

I. Conclusion

This project demonstrates that neural inference can serve as the primary execution model on a minimal RISC-V core. The key findings are:

1. **Descriptor-based neural ISA** provides a clean abstraction for MLP inference, separating firmware setup from hardware execution.
2. **Block-diagonal model composition** enables combining independently trained models without retraining, via a declarative glue JSON format.
3. **OS-like task switching** is achievable through a learned router MLP, demonstrating process scheduling semantics without traditional OS primitives.
4. **Dual-backend validation** ensures correctness across software and hardware implementations through automated differential tests.
5. **Significant speedups** are achievable: up to 2000x over software baselines (C++ backend) and 26.7x over unoptimized RTL (Verilator backend).

The repository contains a reproducible methodology: training pipeline, custom assembler, model compiler, differential

verification, and phase-based benchmarks. This foundation enables further research into neural-first computing architectures.

J. Future Work

1) *Interrupt-Driven Neural Scheduling*: The current execution model is synchronous and loop-driven. Integrating timer and input IRQs would enable event-triggered neural inference, enabling evaluation of bounded latency and preemption behavior.

2) *Dual-Emulator Communication*: Running two emulator instances in lockstep with packet exchange would demonstrate neural multiplayer behavior, introducing synchronization and reproducibility questions for distributed deterministic simulation.

3) *Hardware I/O Abstraction*: Char-device streams and MMIO/register-mapped peripherals with small software drivers would bridge input events to neural tensors and neural outputs to actuator/framebuffer paths.

K. Limitations

This work has several limitations that bound its contributions:

1. **Static model composition**. Block-diagonal merging requires all sub-models to be known at compile time. Adding or removing a sub-model requires regenerating the combined weight matrix and recompiling the firmware. There is no runtime dynamic linking of neural models.
2. **Single-threaded execution**. The inference loop is sequential and cooperative. There is no preemption, no interrupt-driven scheduling, and no parallelism across cores. The OS-like task switching is purely logical---all sub-models execute in the same thread on every frame.
3. **Limited I/O**. Input is restricted to keyboard events via memory-mapped registers. There is no network, storage, or sensor abstraction. The system cannot stream data or communicate with external devices.
4. **MLP-only inference**. The neural ISA extensions target fully connected layers. Convolutional, recurrent, or attention-based architectures are not supported. The hardware FSM is optimized for matrix-vector multiply, not for the data patterns of other layer types.
5. **No traditional OS**. There is no memory protection, no process isolation, and no file system. A misbehaving model can corrupt any memory location. This is acceptable for single-application bare-metal deployment but not for multi-tenant environments.
6. **Deterministic but fragile output**. Bit-exact parity between backends requires identical floating-point semantics. Any platform-specific rounding or compiler optimization could break the differential validation.

These limitations are intentional design choices---the system targets minimal, single-application neural deployment on resource-constrained cores, not general-purpose computing.

L. Acknowledgment

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M. AI Use Disclosure

This paper was produced with the assistance of large language model (LLM) tools. The author used AI tools for code generation, code refactoring, paper writing, figure creation, benchmark analysis, literature review, and reference verification. All AI-generated content was reviewed, verified, and modified by the author. An independent audit using multiple AI agents was conducted to verify technical claims, reference accuracy, and internal consistency. The author takes full responsibility for the content of this paper.

N. Acronyms

- ABI: Application Binary Interface
- CI: Continuous Integration
- CTest: CMake test driver
- ELF: Executable and Linkable Format
- FC: Fully Connected (neural layer)
- ISA: Instruction Set Architecture
- MAC: Multiply-Accumulate
- MLP: Multi-Layer Perceptron
- NRAL: Neural (binary model format magic number)
- PMAC: Parallel Multiply-Accumulate
- RTL: Register Transfer Level

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